Replication materials for "Validating open-source machine translation for quantitative text analysis"

Hauke Licht* Ronja Sczepanski† Moritz Laurer‡ Ayjeren Bekmuratovna§

2026-04-16

Contents

Overview **1**

Replication files **2**
Code . . . 2
Data . . . 2
Tables and figures . . . 2

Reproducibility run **2**
Scope of reproducibility . . . 3
Numerical reproducibility . . . 3
Estimated run times . . . 3

Computer setup **3**
Software requirements . . . 3
R . . . 3
Python . . . 4
Hardware requirements . . . 4

Compute times **4**

Overview

This document describes the replication materials accompanying the **research article**

> Licht, H., Sczepanski, R., Laurer, M. & Bekmuratovna, A (forthcoming). Validating open-source machine translation for quantitative text analysis. *Political Science Research and Methods.*

The replication materials are available in the Political Science Research and Methods (PSRM) Harvard Dataverse[1] at: https://doi.org/10.7910/DVN/8UOOEW

Please address any **questions** you have about the replication materials at the corresponding author Hauke Licht (hauke.licht@ubik.ac.at).

*University of Innsbruck. Corresponding author: hauke.licht@uibk.ac.at
†Scienes Po Paris
‡*Cohere*
§independent researcher
[1]https://dataverse.harvard.edu/dataverse/PSRM

Replication files

This repository contains the code and data for replicating the results reported in the paper and supporting materials. Please refer to the R script `run_replication.R` for a **reproducibility run** of all analyses** (described in the next section).

The replication materials are organized in the following folders:

- `code/`: Contains the code for all experiments and analyses.
- `data/`: Contains the data used all experiments and analyses and quantitative results produced.
- `paper/`: Contains the figures and tables reported in the paper and the supporting materials.

Important To display the materials in their original file structure (instead of a flat list of files), select the **tree view** in the data set files viewer:

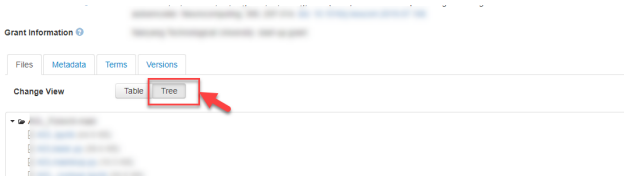

Figure 1: Enable data set files *tree view*

Code

The **code** in this repository is organized into the following folders:

- `code/01-data_preparation/`: R scripts for preparing raw and labeled text data.
- `code/02-machine_translation/`: Python scripts for machine translation of the text data.
- `code/03-topic_modeling/`: R scripts for fitting topic models to the text data.
- `code/04-classifier_finetuning/`: Python scripts for fine-tuning classifiers on the text data.
- `code/05-analyses/`: R scripts for analyzing the results of topic modeling and classifier fine-tuning studies, i.e., Studies I and II.

Data

The **data** in this repository is organized into the following folders:

- `data/exdata/`: External data required for our analyses but not publicly available otherwise.
- `data/datasets`: prepared topic modeling and classifier fine-tuning datasets (based on code in `code/01-data_preparation/`), including texts' translations with open-source machine translation models (based on code in `code/02-machine_translation/`)
- `data/results/`: topic modeling and classifier fine-tuning results

Tables and figures

All **tables and figures** produced in our analyses and reported in the paper and supporting materials can be found in `paper/figures/` and `paper/tables/`, respectively.

Reproducibility run

All data analysis for generating the numbers, tables, and figures reported in the paper and supporting materials were conducted in R. The script `run_replication.R` implements a complete reproducibility run and logs the results to `replication_run.log`.

Scope of reproducibility

The replication run (`run_replication.R`) focuses exclusively on the **main analyses**, i.e., the scripts in `code/05-analyses/`. Data preparation (`code/01-data_preparation/`), topic model fitting (`code/03-topic_modeling/`) and transformer classifier fine-tuning (`code/04-classifier_finetuning/`) are *not* part of the reproducibility run because they require HPC/GPU infrastructure and long compute times (see Section Compute times below). Pre-computed results from these steps are provided in `data/results/` so that the analysis scripts can be run without re-running them.

Numerical reproducibility

The analysis scripts in `code/05-analyses/` read pre-computed results from `data/results/` and produce deterministic outputs. Running `run_replication.R` should therefore reproduce all numbers, tables, and figures in the paper exactly.

Results produced by earlier pipeline steps are subject to the following sources of non-determinism:

- **Machine translation** (`code/02-machine_translation/`): outputs are deterministic given the same model version and hardware, but floating-point non-determinism across GPU architectures may cause minor numerical differences.
- **Topic model fitting** (`code/03-topic_modeling/`): STM models are fit with random initialisation controlled by random number generation state ("seed"); nevertheless, results may differ slightly across runs even with the same seed on different hardware.
- **Classifier fine-tuning** (`code/04-classifier_finetuning/`): transformer fine-tuning on GPUs is non-deterministic by default; exact metric values may differ slightly across runs and GPU types.

Estimated run times

- **Main analyses** (running `run_replication.R` on a consumer-grade laptop): approximately 15 minutes.
- **Full replication**:
  - Machine translation: see Table 1
  - Topic model fitting: approx. 20 minutes with 40 CPU cores on HPC cluster.
  - Classifier fine-tuning: see Tables 2f.

Computer setup

Software requirements

R

We used R for

1. raw and labeled text data preparation (see `code/01-data_preparation`),
2. topics model fitting (see `code/03-topic_modeling`), and
3. analysis of our topic modeling and classifier fine-tuning studies' results (see `code/05-analyses`).

We used R 4.2.0 for the original research. Analyses were successfully reproduced with R 4.5.2.

We use the R package manager `renv` to manage R package dependencies. The `renv` setup and `replication_r_requirements.txt` cover only the packages required for the **main analyses** (`code/05-analyses/`). R packages needed for data preparation and topic model fitting are listed separately in `r_requirements.txt`.

To install all required R packages, open R and run:

```
install.packages("devtools")
devtools::install_version("renv", version = "1.0.10")
renv::restore()
```

A detailed list of R package dependencies and versions can be found in file `replication_r_requirements.txt`

To also replicate the R-based data preparation and topic modeling fitting, please see the additional dependencies in `r_requirements.txt`.

Python

We used Python for

1. machine translation (see `code/02-machine_translation/`), and
2. classifier fine-tuning (see `code/04-classifier_finetuning/`)
3. BERTScore-based translation similarity analysis (see `code/05-analyses/translation_*/`)

We used Python 3.10 and `pip` and the python `venv` module to manage python package dependencies. Specifically, we have kept packages used for machine translation and classifier fine-tuning separate to avoid version conflicts.

To create the virtual environments, open the Terminal app (macOS) or command line (Linux) and run:

```
cd code/02-machine_translation
chmod +x setup_venv.sh
./setup_venv.sh

cd ../03-classifier_finetuning
chmod +x setup_venv.sh
./setup_venv.sh
```

Hardware requirements

All analyses in code/05-analyses/ can be run on a consumer-grade laptop with CPU processors. For machine translation, topic model fitting, and classifier fine-tuning, we used the ETH Zurich's EULER HPC cluster via SLURM. The necessary R, python, and SLURM job scripts can be found in folders `code/02-machine_translation/`, `code/03-topic_modeling/`, and `code/04-classifier_finetuning/`, respectively.

Compute times

Table 1 reports the duration of all machine translation jobs by dataset. Table 2 reports the total duration of all classifier fine-tuning jobs by dataset. Tables 3–7 report the duration of all classifier fine-tuning jobs by dataset and translation source (incl. multilingual training).

Table 1: Duration of translation jobs in hours, minutes, and seconds. *Note:* All jobs run with a single NVIDIA Tesla V100 (40Gb).

Dataset	Duration
Manifesto Translations corpus (Ivanusch & Regel, 2024)	07:14:42
Dupont & Rachuj (2022)	08:19:07
Lehmann & Zobel (2018)	00:45:04
Poljak (2023)	11:30:07
Theocharis et al. (2016)	02:04:01

Table 2: Total runtime of transformer classifier finetuning jobs by dataset in hours, minutes, and seconds. *Notes:* Each jobs run with a single NVIDIA RTX 4090 (24Gb). Number of (No.) translation sources includes 'multilingual.'

Dataset	No. translations	Duration
Dupont & Rachuj (2022)	7	01:22:03
Lehmann & Zobel (2018)	6	00:12:53
Manifesto Translations corpus (Ivanusch & Regel, 2024)	4	01:04:24
Poljak (2023)	7	01:06:56
Theocharis et al. (2016)	7	01:19:58

Table 3: Runtime of transformer classifier finetuning jobs in Dupont & Rachuj (2022) dataset in hours, minutes, and seconds. *Note:* Each jobs run with a single NVIDIA RTX 4090 (24Gb).

		Task				
Translation	Model	topic	rile	rile_binary	econ_position_binary	freedem_position_binary
DeepL	roberta-base	00:02:50	00:04:04	00:02:10	00:00:54	00:00:41
Google Translate	roberta-base	00:02:50	00:04:05	00:02:10	00:00:56	00:00:43
OPUS-MT	roberta-base	00:03:07	00:04:28	00:02:19	00:00:56	00:00:44
M2M (1.2B)	roberta-base	00:03:01	00:04:19	00:02:16	00:00:57	00:00:46
M2M (418M)	roberta-base	00:03:11	00:04:28	00:02:19	00:00:58	00:00:47
multilingual	xlm-roberta-base	00:04:19	00:05:45	00:03:04	00:01:21	00:01:04

Table 4: Runtime of transformer classifier finetuning jobs in Lehmann & Zobel (2018) dataset in hours, minutes, and seconds. *Note:* Each jobs run with a single NVIDIA RTX 4090 (24Gb).

		Task	
Translation	Model	pimpo_issue	pimpo_position
DeepL	roberta-base	00:01:02	00:00:56
Google Translate	roberta-base	00:01:03	00:00:57
OPUS-MT	roberta-base	00:01:03	00:00:58
M2M (1.2B)	roberta-base	00:01:03	00:00:56
M2M (418M)	roberta-base	00:01:03	00:00:57
multilingual	xlm-roberta-base	00:01:31	00:01:24

Table 5: Runtime of transformer classifier finetuning jobs in Manifesto Translations corpus (Ivanusch & Regel, 2024) dataset in hours, minutes, and seconds. *Note:* Each jobs run with a single NVIDIA RTX 4090 (24Gb).

		Task			
Translation	Model	topic	rile	rile_binary	econ_position_binary
DeepL	roberta-base	00:05:49	00:06:26	00:04:20	00:01:56
M2M (1.2B)	roberta-base	00:06:10	00:04:08	00:01:49	00:00:53
M2M (418M)	roberta-base	00:04:37	00:04:27	00:05:42	00:02:32
multilingual	xlm-roberta-base	00:03:16	00:06:16	00:04:10	00:01:53

Table 6: Runtime of transformer classifier finetuning jobs in Poljak (2023) dataset in hours, minutes, and seconds. *Note:* Each jobs run with a single NVIDIA RTX 4090 (24Gb).

		Task		
Translation	Model	attack_binary	attack_type	incivility
DeepL	roberta-base	00:04:34	00:02:31	00:02:16
Google Translate	roberta-base	00:04:34	00:02:30	00:02:16
OPUS-MT	roberta-base	00:04:34	00:02:30	00:02:16
M2M (1.2B)	roberta-base	00:04:35	00:02:30	00:02:16
M2M (418M)	roberta-base	00:04:36	00:02:29	00:02:15
multilingual	xlm-roberta-base	00:05:12	00:03:00	00:02:44

Table 7: Runtime of transformer classifier finetuning jobs in Theocharis et al. (2016) dataset in hours, minutes, and seconds. *Note:* Each jobs run with a single NVIDIA RTX 4090 (24Gb).

		Task				
Translation	Model	sentiment_binary	sentiment	communication	polite	political
DeepL	cardiffnlp/twitter-roberta-base	00:01:22	00:03:19	00:03:18	00:01:06	00:02:59
Google Translate	cardiffnlp/twitter-roberta-base	00:01:18	00:03:11	00:03:09	00:01:04	00:02:52
OPUS-MT	cardiffnlp/twitter-roberta-base	00:01:29	00:03:33	00:03:32	00:01:14	00:03:15
M2M (1.2B)	cardiffnlp/twitter-roberta-base	00:01:20	00:03:18	00:03:18	00:01:05	00:02:57
M2M (418M)	cardiffnlp/twitter-roberta-base	00:01:35	00:04:03	00:04:04	00:01:20	00:03:38
multilingual	cardiffnlp/twitter-xlm-roberta-base	00:01:56	00:04:22	00:04:19	00:01:30	00:03:54